Task 3: Modelling Budgets and Downside Risk

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1 Task 3: Modelling Budgets and Downside Risk

1.1 Introduction

Meridian's Finance team needs two inputs for planning: a realistic probability model for production budgets, benchmarked against Universal Pictures, for its budget simulator; and a downside-risk figure for Action, its most capital-intensive genre. Downside risk is the 25th percentile of ROI (worldwide box-office revenue divided by production budget), the multiple below which the worst quarter of Action films fall.

1.2 Universal Pictures Budget Modelling

We used the 243 Universal Pictures films (by `primary_company`) released up to 2016 with a valid budget, treating budgets under $10,000 as missing-data placeholders. Budgets (in $m) are strongly right-skewed: the middle half lie between $17m and $60.5m, but a few exceed $150m, pulling the mean ($44.5m) well above the median ($35m). Alongside the required normal distribution, we fitted gamma, log-normal and Weibull distributions, which take only positive values and allow for right skew, all by maximum likelihood using `fitdistr()` from the MASS package.

1.3 Model Diagnostics and Comparison

Distribution	Parameters (units)	Estimate 1	Estimate 2	Log-likelihood	AIC
Gamma	shape, rate (per $m)	1.382	0.031	-1,158.1	2,320.1
Weibull	shape, scale ($m)	1.214	47.479	-1,158.3	2,320.6
Log-normal	meanlog, sdlog (log $m)	3.392	1.015	-1,172.6	2,349.2
Normal	mean, sd ($m)	44.481	37.432	-1,225.1	2,454.1

Table 1: Distributions fitted to Universal Pictures budgets, ordered by AIC (lowest first)

AIC rewards a close fit but penalises extra parameters, so lower is better. As all four distributions have two parameters, the ranking in Table 1 reflects fit alone. The gamma has the lowest AIC (2320.1), with the Weibull effectively tied (2320.6); the log-normal is 29 points worse and the normal is far behind (2454.1). AIC cannot separate the top two or show whether either fits well, so we also used Q-Q plots.

In Figure 1, the normal curves away at both ends: its lowest theoretical quantiles are negative (it gives about a 12% chance of a negative budget), and the largest budgets sit well above the line, so it understates expensive films. The log-normal errs the other way: its top theoretical quantile is over $500m, against Universal's actual maximum of $209m. The gamma and Weibull both track the line, but for the biggest budgets the Weibull points drift above it while the gamma stays close. As large budgets drive Finance's exposure, and the gamma also has the lowest AIC, we recommend the gamma distribution (shape 1.382, rate 0.031 per $m).

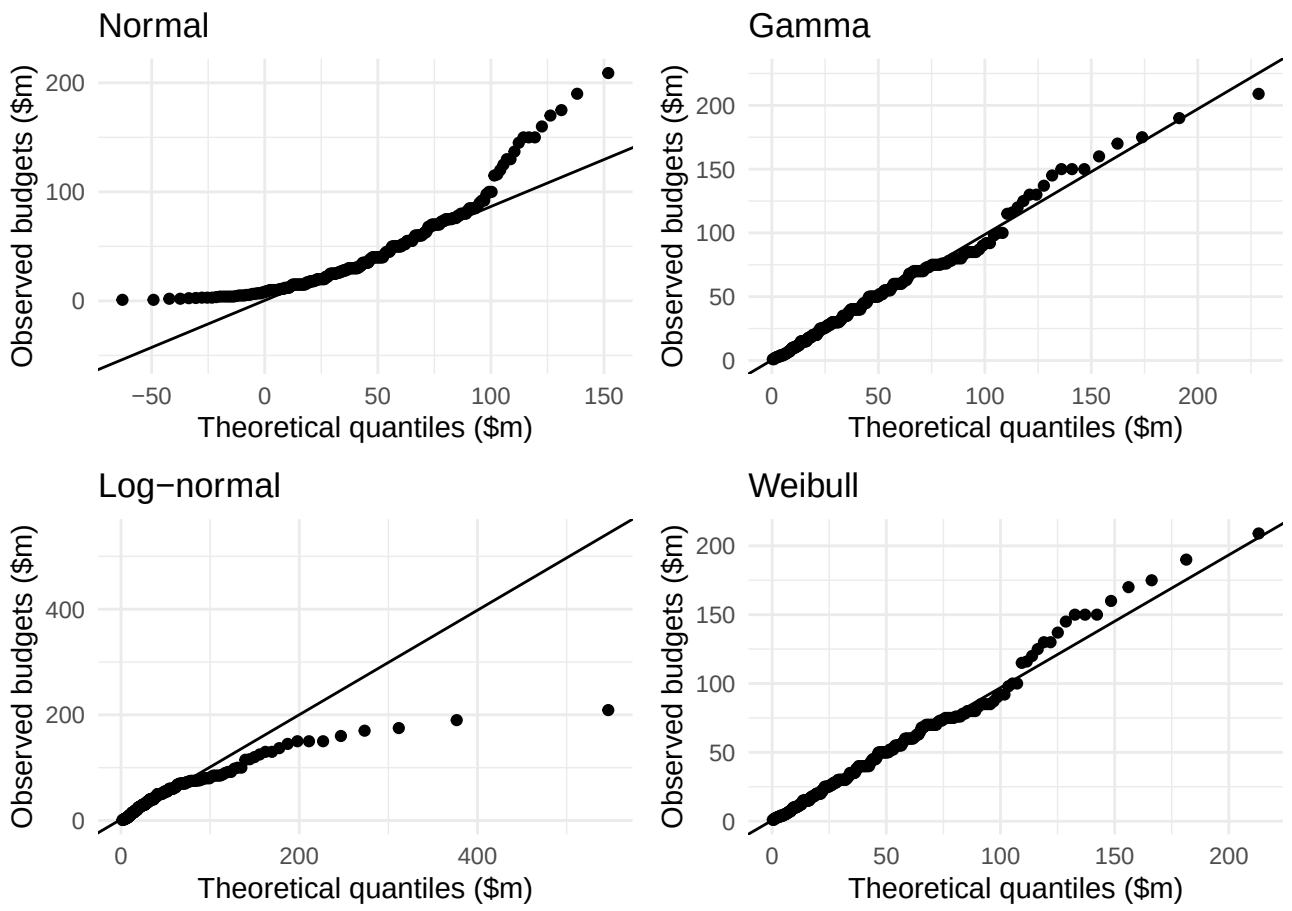


Figure 1: Q-Q plots of Universal Pictures budgets against each fitted distribution. Points close to the line indicate a good fit.

1.4 Action Genre Downside Risk

Method	25th percentile of ROI	95% lower	95% upper
Percentile bootstrap	0.91	0.80	1.05
Fitted log-normal model	0.73	0.66	0.82

Table 2: 95% interval estimates for the 25th percentile of ROI for Action films, from two methods

1.5 Comparison of Interval Estimates

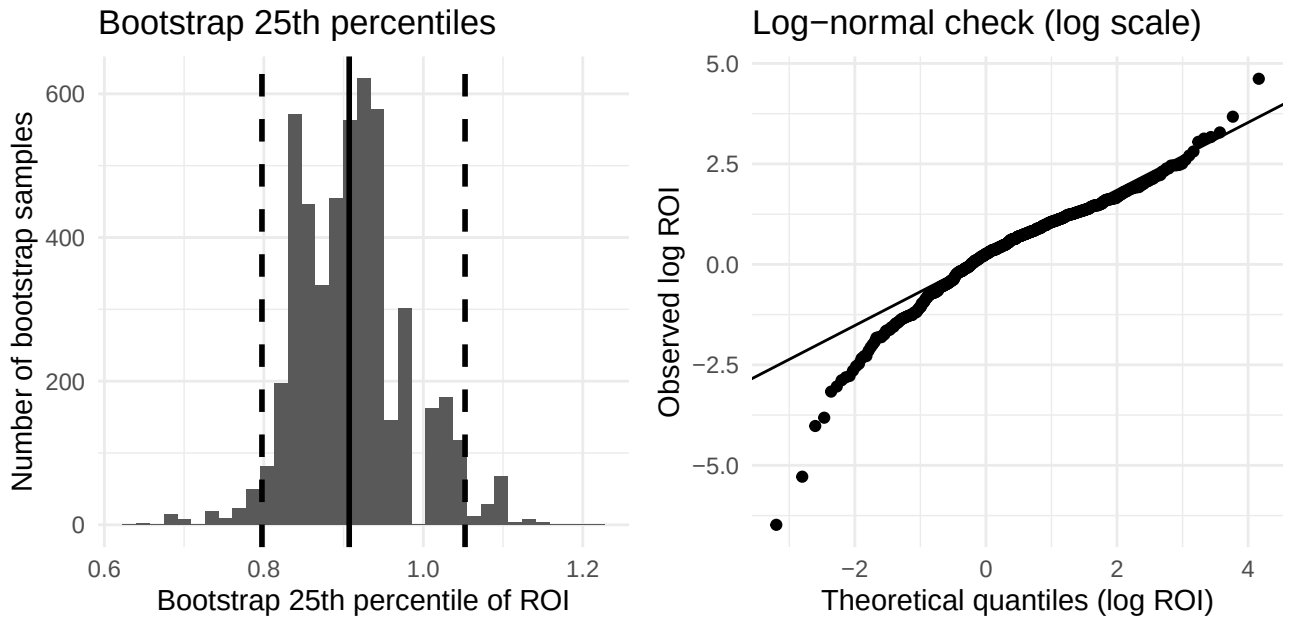


Figure 2: Left: bootstrap distribution of the 25th percentile of Action-film ROI (solid line: sample estimate; dashed lines: 95% interval). Right: Q-Q plot of log ROI against the fitted log-normal model.

1.6 Summary for the Finance Team